Independent Thermoelectric Node Regulation for Adaptive Rooftop Snow Mitigation Under Cold Climate Conditions

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***Abstract*—This paper presents an independent thermoelectric node regulation framework for adaptive rooftop snow mitigation under cold climate conditions. A distributed thermoelectric grid composed of independently controlled thermal nodes is proposed for localized rooftop thermal stabilization using Seebeck-based thermal-electrical regulation.**

A transient thermal simulation framework is developed to analyze rooftop heat diffusion, adaptive thermal control, and thermal stabilization under winter environmental conditions representative of Yamagata, Japan. The proposed architecture compares classical PID regulation against adaptive PID control combined with Kalman-based thermal state estimation.

Simulation results demonstrate improved thermal stabilization, reduced overshoot, and enhanced robustness under adaptive local regulation. The proposed framework establishes a foundation for future intelligent thermoelectric infrastructures, digital twin integration, and AI-assisted thermal control systems.

I. Introduction

Snow accumulation on residential rooftops represents a major engineering challenge in cold-weather regions due to structural loading, thermal inefficiencies, and safety risks associated with ice formation. Conventional rooftop snow mitigation systems typically rely on resistive heating or manual intervention, both of which may present limitations in scalability and energy efficiency [6], [9].

Thermoelectric generators (TEGs) provide an alternative approach for localized thermal regulation through distributed thermal-electrical conversion mechanisms based on the Seebeck effect [1], [2], [8]. By exploiting temperature gradients across thermoelectric materials, independently controlled thermoelectric nodes may dynamically regulate rooftop thermal conditions while minimizing unnecessary energy consumption.

This paper proposes an independent thermoelectric node regulation framework for adaptive rooftop snow mitigation under winter environmental conditions representative of Yamagata, Japan. The proposed framework models a spatial thermoelectric grid composed of independently regulated thermal nodes embedded beneath rooftop surfaces.

The main contributions of this work include the development of an independent thermoelectric rooftop node regulation architecture, localized thermal interaction modeling based on Seebeck thermal behavior, rooftop snow mitigation simulation under realistic winter conditions, comparative evaluation between ON/OFF, PID, and adaptive PID regulation, and a future digital twin framework for AI-assisted thermal regulation.

II. Related Work

Previous research on thermoelectric systems has primarily focused on waste heat recovery, industrial thermal management, and low-power energy harvesting applications [1], [2], [8], [9]. Several studies have investigated transient thermal modeling and adaptive thermal regulation in thermoelectric systems under dynamic environmental conditions [7], [12].

Recent work has also explored smart thermal infrastructures for cold-weather environments, including electrically heated rooftop systems and distributed thermal sensing architectures [6], [13]. However, most conventional rooftop snow mitigation systems still rely on resistive heating methods with limited adaptive regulation capability.

Research on distributed thermoelectric rooftop infrastructures remains limited, particularly regarding independently regulated thermal node architectures under transient environmental conditions. Additionally, adaptive thermal regulation strategies combining PID control and Kalman-based thermal estimation have not been extensively investigated for intelligent snow mitigation infrastructures [3]–[5], [13].

The proposed work contributes by integrating thermoelectric regulation, thermal diffusion modeling, adaptive control, and future AI-assisted digital twin integration within a unified simulation framework.

III. Independent Thermoelectric Rooftop Architecture

A. Thermoelectric Node Grid

The proposed system consists of a spatial grid of independently regulated thermoelectric nodes embedded beneath rooftop surfaces. Each thermoelectric node contains:

- thermoelectric module,

- local thermal sensor,
- voltage and current monitoring,
- adaptive thermal controller,
- battery-assisted regulation subsystem.

Each node independently regulates localized rooftop temperature while thermally interacting with neighboring nodes through passive thermal diffusion.

Figure 1 illustrates the proposed distributed thermoelectric rooftop architecture.

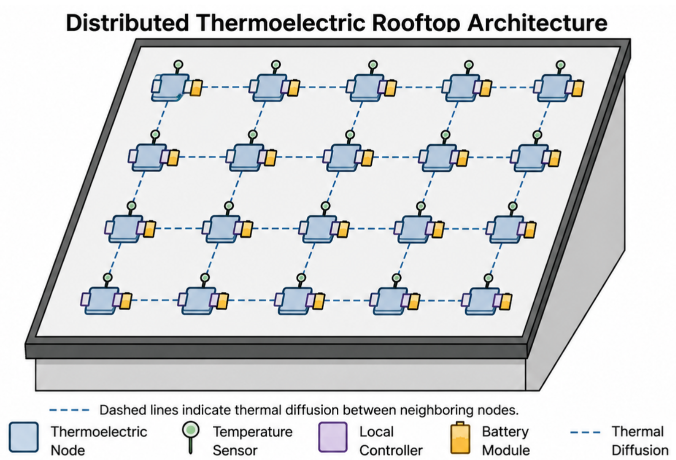

Fig. 1: Distributed thermoelectric rooftop node architecture for adaptive snow mitigation using independently regulated thermal nodes.

B. Thermoelectric Model

The overall thermoelectric figure of merit is defined as:

$$ZT = \frac{S^2 \sigma T}{k} \tag{1}$$

where S is the Seebeck coefficient, σ is electrical conductivity, T is absolute temperature, and k is thermal conductivity [1], [8].

The thermal-electric conversion efficiency is approximated by:

$$\eta = \frac{\Delta T}{T_h} \cdot \frac{\sqrt{1+ZT}-1}{\sqrt{1+ZT}+\frac{T_c}{T_h}} \tag{2}$$

The voltage generated at each thermoelectric node is modeled according to the Seebeck effect:

$$V_i = \alpha(T_{h,i} - T_{c,i}) \tag{3}$$

where α is the Seebeck coefficient.

The transient thermal dynamics of each node are modeled as:

$$C_i \frac{dT_i}{dt} = Q_{local,i} - Q_{loss,i} + Q_{neighbor,i} \tag{4}$$

Thermal propagation across the rooftop surface is modeled using a discrete thermal diffusion approximation [6], [7]:

$$T_{i,j}^{t+1} = T_{i,j}^{t} + \Delta t \left(Q_{local} - Q_{loss} + k \nabla^2 T \right) \tag{5}$$

C. Environmental Conditions

The simulation environment considers winter conditions representative of Yamagata, Japan, including ambient temperatures ranging from $-8°C$ to $2°C$, rooftop thermal fluctuations, snow accumulation, wind-induced thermal losses, and transient environmental perturbations.

These operating conditions were selected based on winter climate statistics reported by the Japan Meteorological Agency and environmental reports from Yamagata Prefecture [10], [11].

IV. Local Thermal Control Architecture

The proposed control system independently regulates each thermoelectric node to maintain rooftop temperatures above snow accumulation thresholds while minimizing energy consumption.

Each node evaluates:

- local thermal state,
- local temperature gradient,
- battery charge availability,
- estimated snow accumulation,
- local thermal dissipation.

The adaptive thermal regulation framework combines localized PID-based thermal control with Kalman filter thermal state estimation for improved robustness under environmental uncertainty and sensor noise conditions.

A proportional-integral-derivative (PID) controller regulates local thermal response according to [3]:

$$u_i(t) = K_p e_i(t) + K_i \int e_i(t)dt + K_d \frac{de_i(t)}{dt} \tag{6}$$

where $u_i(t)$ is the control signal and $e_i(t)$ is the thermal regulation error.

To improve robustness under environmental uncertainty, adaptive thermal regulation is implemented using Kalman filter-based thermal state estimation [4], [5]. The adaptive controller dynamically adjusts proportional gain values according to the estimated thermal error magnitude, allowing smoother convergence under transient environmental perturbations.

V. Simulation Methodology

Table I summarizes the primary simulation parameters used throughout the thermal analysis framework.

TABLE I: Simulation Parameters

Parameter	Value
Grid Size	10×10 nodes
Ambient Temperature	$-8°C$ to $2°C$
Target Temperature	$2°C$
Simulation Step	0.1 s
Thermal Diffusion Coefficient	0.15
Seebeck Coefficient	200 $\mu V/K$
PID Gains (K_p, K_i, K_d)	$2.5, 0.12, 0.55$
Adaptive PID Gains	$2.2, 0.09, 0.65$
Kalman Process Noise	0.01
Kalman Measurement Noise	0.08

The rooftop surface is modeled as a two-dimensional thermal grid where each node exchanges thermal energy with adjacent nodes through passive thermal diffusion.

The simulation framework considers independent node operation, environmental perturbations, snow accumulation dynamics, localized adaptive regulation, and battery-assisted thermal stabilization.

Simulation scenarios include rapid ambient temperature drops, intermittent snow accumulation, partial node failures, and localized adaptive thermal response.

VI. Simulation Results

A. Distributed Thermal Distribution

The proposed thermal diffusion model generated stable spatial thermal regulation across the rooftop surface under severe winter operating conditions.

Figure 2 illustrates the steady-state rooftop thermal distribution obtained from the distributed thermoelectric node simulation.

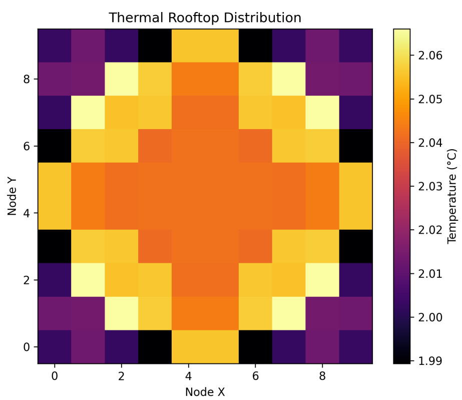

Fig. 2: Thermal distribution generated by independently regulated thermoelectric rooftop nodes during adaptive snow mitigation operation.

B. Transient Thermal Response

Figure 3 presents the transient thermal response of a representative thermoelectric node under adaptive thermal regulation.

The obtained response demonstrates rapid thermal stabilization with low steady-state oscillation around the desired rooftop temperature threshold.

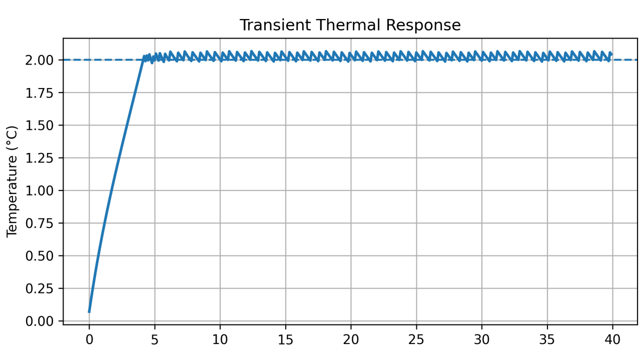

Fig. 3: Transient thermal stabilization response under environmental perturbations.

C. Kalman Thermal State Estimation

Figure 4 illustrates the thermal state estimation process using a Kalman filter under noisy environmental measurements.

The Kalman-based estimator smooths thermal sensor noise and improves thermal state observability during transient environmental perturbations.

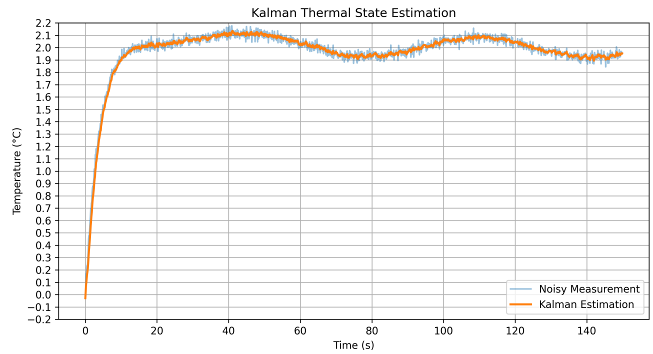

Fig. 4: Kalman filter-based thermal state estimation under noisy environmental measurements.

D. Adaptive PID Thermal Regulation

Figure 5 compares the transient thermal response between a classical PID controller and the proposed adaptive PID controller combined with Kalman thermal estimation.

The adaptive controller achieved improved convergence toward the target thermal state with reduced oscillation amplitude and lower overshoot compared with classical PID regulation.

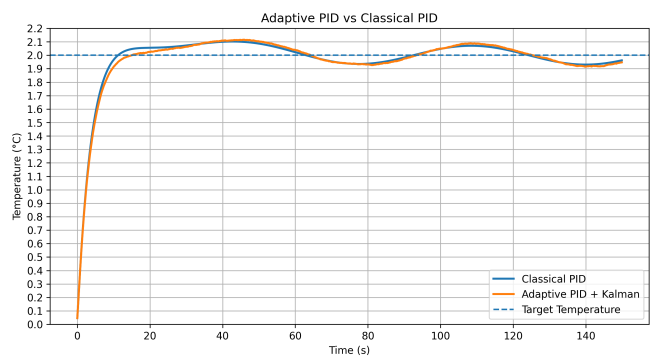

Fig. 5: Comparison between classical PID regulation and adaptive PID with Kalman-based thermal estimation.

E. Controller Benchmark Comparison

Figure 6 compares conventional ON/OFF thermal regulation against PID-based thermal regulation.

The ON/OFF controller exhibits larger oscillatory behavior and slower convergence toward the target rooftop temperature. In contrast, PID regulation achieves smoother thermal stabilization with lower steady-state oscillation and improved thermal convergence performance.

Table II summarizes the comparative thermal control performance metrics obtained from the simulation benchmark.

The obtained results indicate that adaptive thermal regulation combined with Kalman-based estimation improves overall

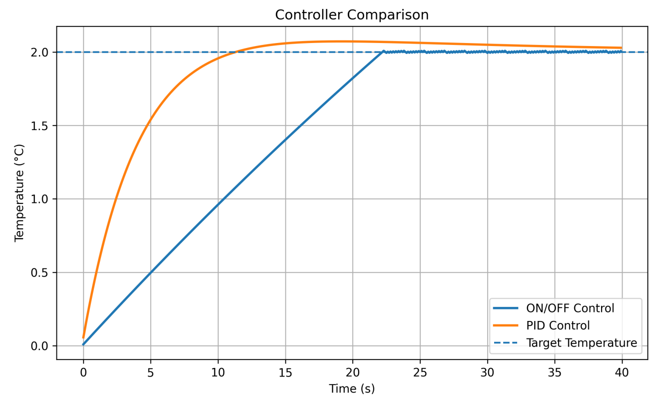

Fig. 6: Comparison between conventional ON/OFF thermal regulation and PID-based thermal control.

TABLE II: Thermal Control Performance Comparison

Metric	ON/OFF	Adaptive PID
Convergence Speed	Moderate	Fast
Thermal Overshoot	Moderate	Low
Steady-State Oscillation	High	Reduced
Noise Sensitivity	High	Reduced
Thermal Stability	Moderate	High

thermal stabilization performance under uncertain environmental conditions.

Compared with conventional ON/OFF regulation, The proposed framework improves convergence and thermal stability under noisy environmental conditions.

F. Fault-Tolerant Thermal Regulation

Additional simulations were performed under random thermoelectric node failure conditions.

The proposed control architecture preserved overall rooftop thermal stability despite partial node failures.

Figure 7 compares nominal operation against adaptive fault-tolerant recovery behavior.

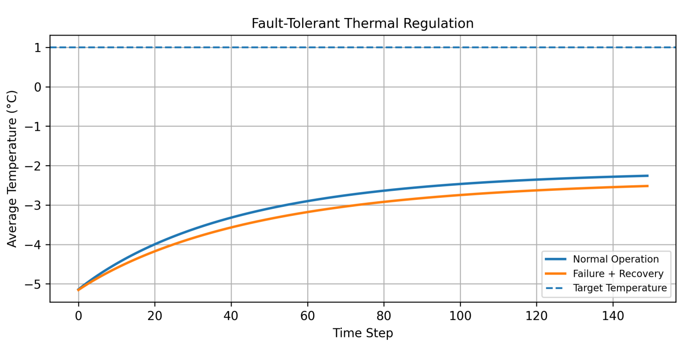

Fig. 7: Fault-tolerant thermal response.

Figure 8 illustrates distributed thermal regulation under partial node degradation and adaptive neighboring thermal compensation.

VII. Future Work

Future work will investigate digital twin integration for real-time rooftop thermal monitoring and predictive thermal regulation under dynamic winter environmental conditions.

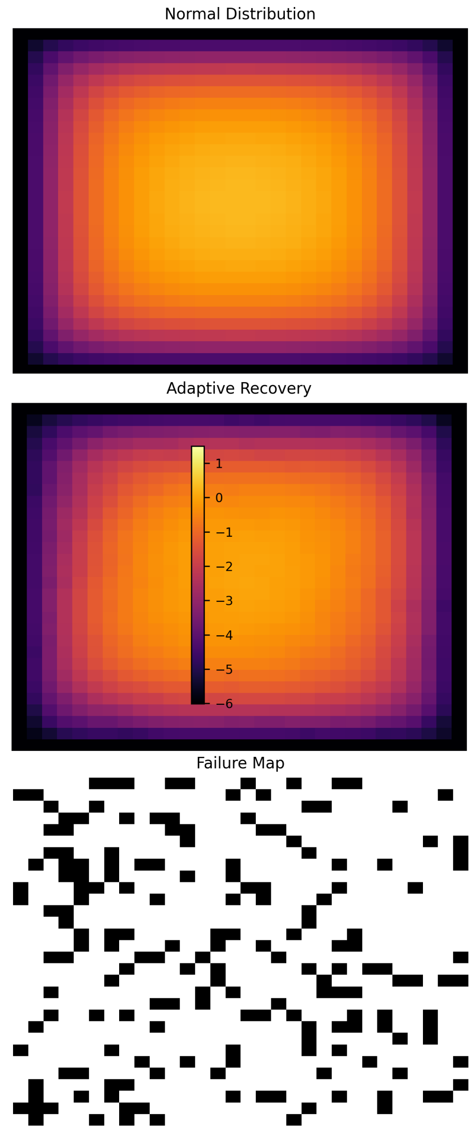

Fig. 8: Fault-tolerant distributed thermal regulation.

Additional research will evaluate AI-assisted thermal prediction and autonomous distributed thermal optimization for large-scale thermoelectric rooftop infrastructures.

VIII. Conclusion

This paper presented a distributed thermoelectric rooftop regulation framework for adaptive snow mitigation under cold-weather conditions.

The proposed system combined localized thermoelectric regulation, thermal diffusion modeling, adaptive PID control, Kalman-based thermal estimation, and fault-tolerant thermal compensation.

Simulation results demonstrated stable rooftop thermal regulation, improved disturbance robustness, and successful operation under partial node failure conditions.

The obtained results support the feasibility of distributed thermoelectric infrastructures for future intelligent snow mitigation systems.

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